Final Project Report

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Project Name - **Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report**

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1. Introduction

1.1. Project Overview

The project, **"Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report"**, is a Business Intelligence and data visualization project designed to study the adoption, usage, and growth of Artificial Intelligence in the global education sector.

The project uses a structured **Global AI in Education** dataset containing various attributes related to AI adoption in education. These include countries, regions, years, AI usage levels, student and teacher AI usage, school AI adoption, rural and urban AI usage, AI tools, government AI policies, and AI adoption levels.

The data is explored and prepared before being imported into **Tableau**. Tableau is then used to create charts, graphs, geographic maps, comparative visualizations, dashboards, and a story that communicate important insights about the global adoption of AI in education.

The project focuses on three major perspectives:

Global Education Perspective

The analysis provides a global view of how Artificial Intelligence is being adopted across different countries and regions. It helps understand geographical patterns and differences in AI adoption within the education sector.

Student and Teacher Perspective

Students and teachers are important users of AI in education. The analysis compares their AI usage levels and helps identify how AI adoption has developed among these two groups over time.

School and Community Perspective

The project also examines AI adoption at the school and community level. It analyses school AI adoption across different regions and compares AI usage between **urban and rural communities** to understand differences in adoption patterns.

Overall, the project provides a **data-driven view of global AI adoption in education** and supports better understanding of AI usage patterns, adoption trends, regional differences, and the growing role of AI in the education sector.

1.2. Objectives

The major objectives of the project are:

- To analyze **global AI adoption in the education sector**.
- To understand the distribution of AI adoption across different **countries and regions**.
- To analyze **average daily AI usage** across different countries.
- To compare **student and teacher AI usage** across different countries.
- To study the growth of **student, teacher, and school AI adoption over the years 2014–2026**.
- To identify and compare the usage of **leading AI tools** in education.

- To analyze the differences in AI usage between **urban and rural communities**.
- To compare **school AI adoption across different geographical regions**.
- To classify educational AI adoption into **High, Moderate, and Low Adoption** levels.
- To understand the geographical distribution of AI adoption using **interactive maps**.
- To create meaningful and interactive visualizations using **Tableau**.
- To develop an interactive dashboard for **global education and AI adoption insights**.
- To present the complete analysis through a **Tableau story**.
- To support better understanding of AI adoption trends in education using **data-driven insights**.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

The rapid growth of **Artificial Intelligence in the education sector** has created new opportunities for improving teaching, learning, assessment, and educational management. However, the adoption of AI varies significantly across countries, regions, educational institutions, and communities.

Students, teachers, and educational institutions may have different levels of AI usage and adoption. It can be difficult to understand which countries have higher AI usage, how student and teacher adoption is changing over time, which AI tools are being used more frequently, and how adoption differs between urban and rural communities.

Educational institutions and decision-makers also need to understand regional differences in school AI adoption and identify whether the overall level of AI adoption is **High, Moderate, or Low**.

Therefore, there is a need for a **Business Intelligence solution** that can analyze global AI adoption in education and present meaningful insights through simple and interactive visualizations.

The proposed project uses **Tableau** to transform the available educational AI data into interactive charts, maps, dashboards, and analytical views, helping users understand global AI adoption patterns and trends in the education sector.

2.2. Project Proposal (Proposed Solution)

The proposed solution is to develop an interactive **Global AI Adoption in Education Dashboard** using Tableau.

The project will follow a data analytics workflow in which the Global AI in Education data is collected, examined, prepared, and visualized to identify meaningful patterns and trends.

The proposed system will provide:

- **Geographic analysis of AI adoption in education**
- **Country-level average daily AI usage analysis**
- **Student AI usage analysis**
- **Teacher AI usage analysis**
- **Student vs Teacher AI usage comparison**
- **AI adoption trend analysis from 2014–2026**
- **Leading AI tools comparison**
- **Urban vs Rural AI usage analysis**
- **Regional school AI adoption analysis**
- **AI adoption level analysis**
- **Interactive visualizations**
- **Charts, graphs, and maps**
- **Dashboard-based insights**
- **Tableau story for presenting findings**

The dashboard will allow users to analyse AI adoption according to different parameters such as **country, region, year, AI usage, educational group, community type, and AI adoption level**.

The final solution will convert complex global education and AI adoption data into an **easy-to-understand visual format** that can help users understand global trends, regional differences, and the growing adoption of Artificial Intelligence in education.

2.3. Initial Project Planning

The proposed solution is to develop an interactive **Global AI Adoption in Education Dashboard** using Tableau.

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- Geographic analysis of AI adoption in education

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- AI adoption level analysis
- Interactive visualizations
- Charts, graphs, and maps
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The dashboard will allow users to analyse and compare the data according to different parameters such as **country, region, year, student usage, teacher usage, school adoption, community type, and AI adoption level**.

The final solution will convert complex global education and AI adoption data into an **easy-to-understand visual format** that can support data-driven understanding of AI adoption trends, regional differences, and educational AI usage patterns.

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The project requires structured data related to **Global AI Adoption in Education**.

The dataset contains information about different dimensions of AI adoption in education, including:

- Country
- Region
- Year
- Month
- AI in Curriculum
- Government AI Policy
- Average Daily AI Usage
- Student AI Usage Percentage
- Teacher AI Usage Percentage
- School AI Adoption
- Rural AI Usage Percentage
- Urban AI Usage Percentage
- Gender Gap in AI Usage

- Internet Penetration
- AI Adoption Level
- Top AI Tool

The raw dataset was identified as the primary source for the analysis. The data was obtained in a structured format suitable for analysis and visualization using Tableau.

Data Collection Process

1. Identify the requirements of the project.
2. Identify relevant global AI-in-education data.
3. Obtain the dataset in a structured format.
4. Examine the available columns and records.
5. Identify missing or inconsistent values.
6. Prepare the dataset for analysis.
7. Import the dataset into Tableau.
8. Verify the data before creating visualizations.

Expected Data Dimensions

The project primarily analyses AI adoption in education according to:

Country → Region → Year → AI Usage → Student/Teacher Usage → School Adoption → AI Tool → Adoption Level

These dimensions provide multiple perspectives for understanding the global adoption and usage of Artificial Intelligence in education.

3.2. Data Quality Report

Before developing visualizations, the dataset was examined to identify potential quality issues and ensure that the available data was suitable for analysis.

The following data-quality checks were performed:

1. Missing Values

The dataset was checked for blank or null values in important fields such as country, region, year, AI usage, student usage, teacher usage, and school adoption.

2. Duplicate Records

Records were examined for possible duplication to ensure that the same educational AI information was not counted multiple times during analysis.

3. Data Consistency

Categorical values such as country, region, AI adoption level, and top AI tool were checked for consistency. Similar categories with different spellings or formats were identified and standardized where required.

4. Data Type Validation

Numerical fields such as average daily AI usage, student AI usage percentage, teacher AI usage percentage, rural and urban usage percentages, and internet penetration were checked to ensure that they were stored as numerical values. Categorical fields were appropriately treated as dimensions.

5. Year and Date Validation

Year and month-related fields were reviewed to ensure that the time-based information was suitable for analysing AI adoption trends from **2014 to 2026**.

6. Category Validation

Country, region, AI adoption level, top AI tool, and other categorical fields were reviewed for consistency before creating Tableau visualizations.

3.3 Data Exploration and Preprocessing

Data exploration was performed to understand the structure and characteristics of the Global AI in Education dataset.

The following activities were performed:

1. Examining the number of records and columns.
2. Identifying categorical and numerical fields.
3. Checking missing values.
4. Checking duplicate records.
5. Reviewing unique values in important dimensions such as country, region, year, AI adoption level, and AI tools.
6. Standardizing inconsistent categorical values where required.
7. Checking AI usage-related information such as average daily AI usage and usage percentages.
8. Reviewing student and teacher AI usage values.
9. Reviewing school, urban, and rural AI adoption values.
10. Preparing the dataset for Tableau visualization.

Data Preprocessing

The preprocessing process involved:

- Removing unnecessary fields where applicable.
- Handling missing values.
- Removing duplicate records where applicable.
- Standardizing categorical values.
- Correcting data types.
- Preparing calculated fields where required.
- Organizing fields for effective Tableau visualization.
- Preparing country and regional data for geographic analysis.
- Organizing year-wise data for AI adoption trend analysis.

After preprocessing, the dataset was imported into Tableau for analysis and visualization.

4. Data Visualization

4.1 Framing Business Questions

Business questions were developed to convert the raw **Global AI in Education** dataset into meaningful analytical objectives.

The major questions considered in the project were:

General AI Adoption Questions

1. What is the geographical distribution of AI adoption in education across different countries?
2. Which countries have higher average daily AI usage?
3. How does AI usage differ between students and teachers?
4. How has AI adoption in education changed from 2014 to 2026?
5. Which AI tools are most commonly used in education?
6. What is the overall distribution of AI adoption levels?

Student and Teacher Questions

1. What percentage of students are using AI in education?
2. What percentage of teachers are using AI?
3. How does student AI usage compare with teacher AI usage?
4. How has student AI usage changed over time?

5. How has teacher AI usage changed over time?

School and Community Questions

1. Which regions have higher levels of school AI adoption?
2. How does AI usage differ between urban and rural communities?
3. How has urban AI usage changed over time?
4. How has rural AI usage changed over time?
5. Which geographical regions show stronger school-level AI adoption?

AI Tool and Adoption Questions

1. Which AI tools are most widely used in education?
2. How are ChatGPT, Google Gemini, Khanmigo, and Microsoft Copilot represented in the dataset?
3. What proportion of the dataset falls under High, Moderate, and Low AI Adoption levels?
4. Which countries and regions show stronger AI adoption?
5. What major patterns can be identified from the global AI adoption data?

These business questions provide the foundation for developing the Tableau visualizations and help convert the dataset into meaningful insights about **global AI adoption in education**.

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4.2. Developing Visualizations

Visualization 1: Geographic Trends in Education Development

A **geographic map visualization** was developed to represent the geographical distribution of AI adoption and education-related data across different countries.

The visualization uses:

- **Country** as the geographical dimension.
- **AI adoption and education-related data** to represent the country-level information.
- **Country names** to identify the different locations included in the dataset.

The map provides a global view of the countries covered in the analysis and helps users understand the geographical scope of AI adoption in education.

The visualization includes countries such as **Australia, Brazil, Canada, China, Germany, India, Nigeria, Pakistan, the United Kingdom, and the United States**, among others.

Purpose:
This visualization helps users understand the geographical distribution of the dataset and provides a foundation for comparing AI adoption patterns across different countries.

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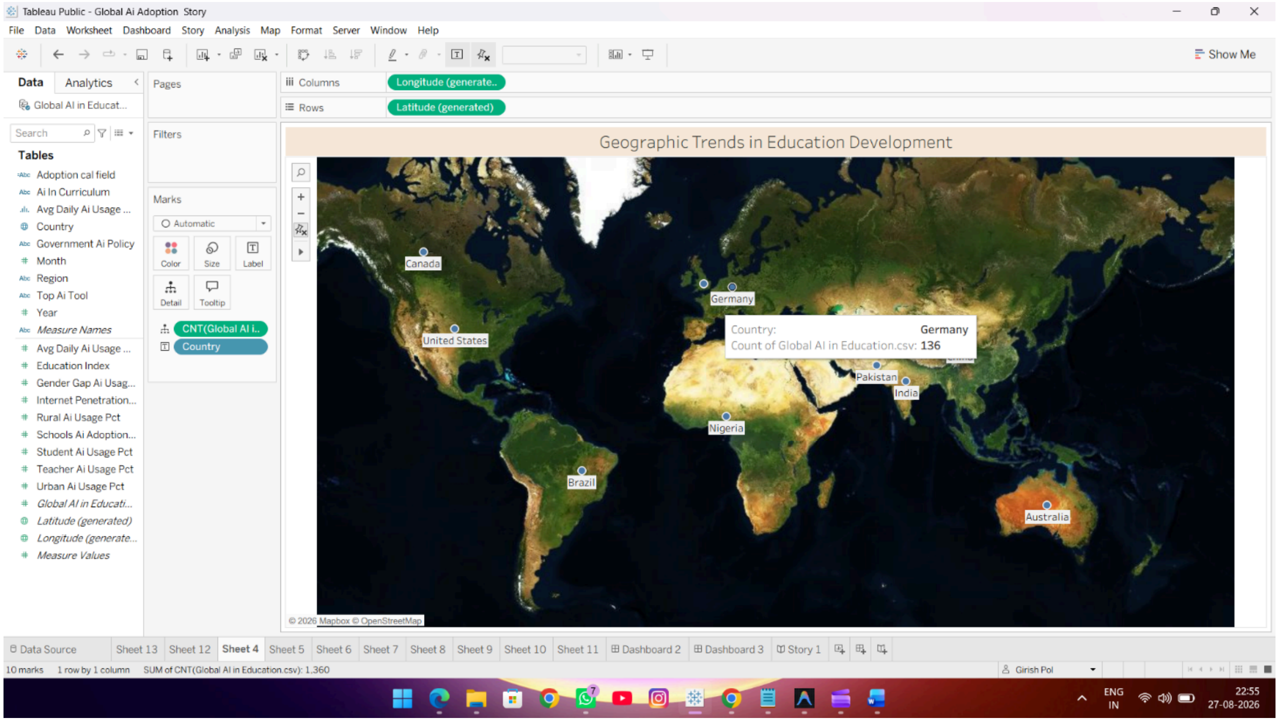


Visualization 2: Country Level AI Adoption in Daily Practice

A **horizontal bar chart** was developed to compare the average daily AI usage across different countries.

The visualization uses:

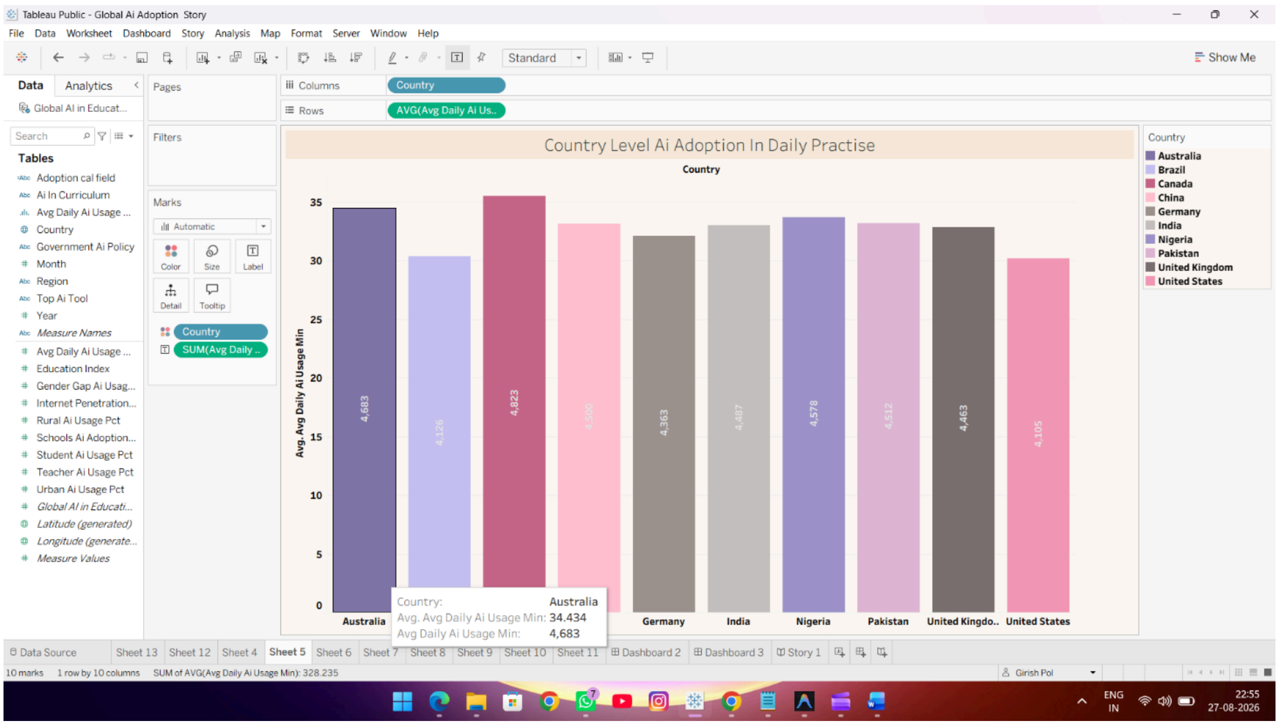
- **Country** as the primary categorical dimension.
- **Average Daily AI Usage** as the measure for comparison.
- Country-wise AI usage values to identify differences in daily AI adoption.

The chart compares the average daily AI usage of different countries included in the dataset. It provides a clear view of how frequently AI is used in educational practices across countries.

The visualization shows differences in daily AI usage among the selected countries, helping identify countries with relatively higher and lower levels of AI usage.

Purpose:

This visualization helps users understand **country-level differences in daily AI usage** and provides a comparative view of AI adoption practices in education across different countries.



Visualization 3: Student vs Teacher AI Usage

A **comparative bar chart** was created to analyse and compare AI usage between students and teachers.

The visualization contains the following categories:

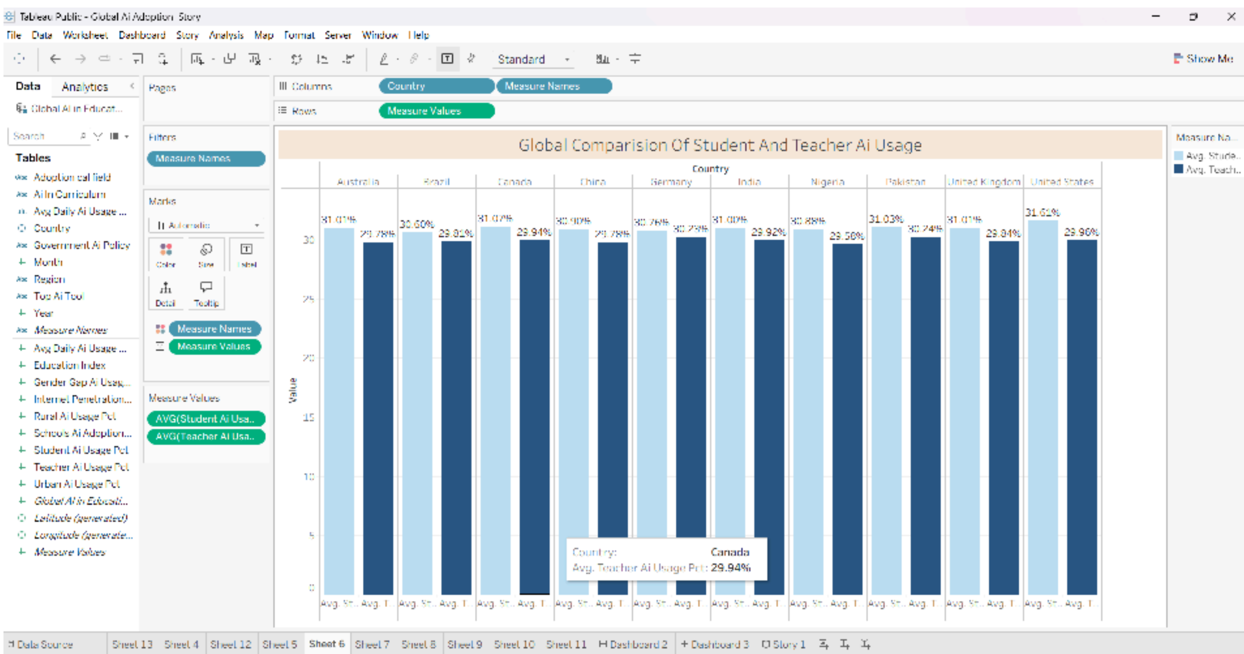
- **Students**
- **Teachers**

The corresponding **AI Usage Percentage** is used to compare the level of AI usage between the two educational groups.

The visualization provides a clear comparison of how frequently students and teachers are using Artificial Intelligence in education. It helps identify the difference between student-level and teacher-level AI adoption.

Purpose:

This visualization helps users compare **AI usage among students and teachers** and understand the adoption of Artificial Intelligence from both learner and educator perspectives.

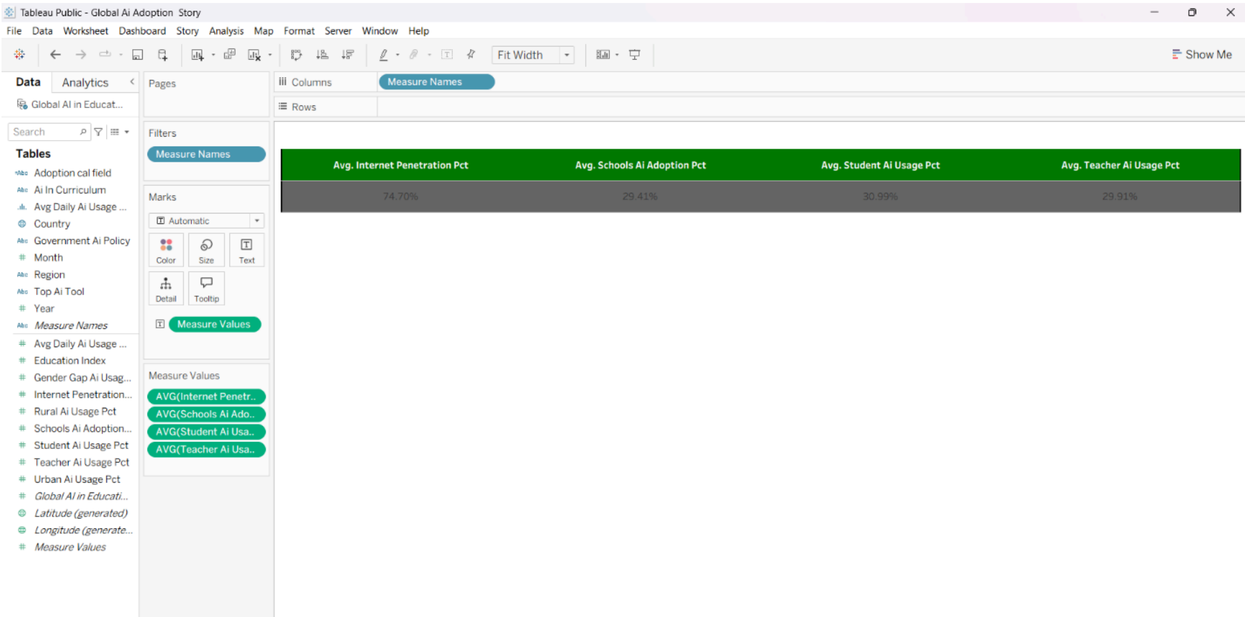


Visualization 4: Key Performance Indicator (KPI)

A **KPI visualization** was created to provide a quick summary of important numerical indicators related to AI adoption in education.The KPI section contains key metrics related to:

Average Daily AI Usage
Student AI Usage Percentage

Teacher AI Usage Percentage



These metrics provide a high-level summary of AI usage in the education sector before users explore the detailed visualizations and comparisons.The KPI visualization gives users an immediate overview of the overall level of AI usage among the major educational groups.

Purpose:

The KPI visualization allows users to understand the **overall AI adoption and usage characteristics of the education sector at a glance** and provides a quick starting point for further analysis.

Visualization 5: AI Tool Usage Comparison

A **horizontal bar chart** was developed to compare the usage of different AI tools in the education sector.

The visualization uses:

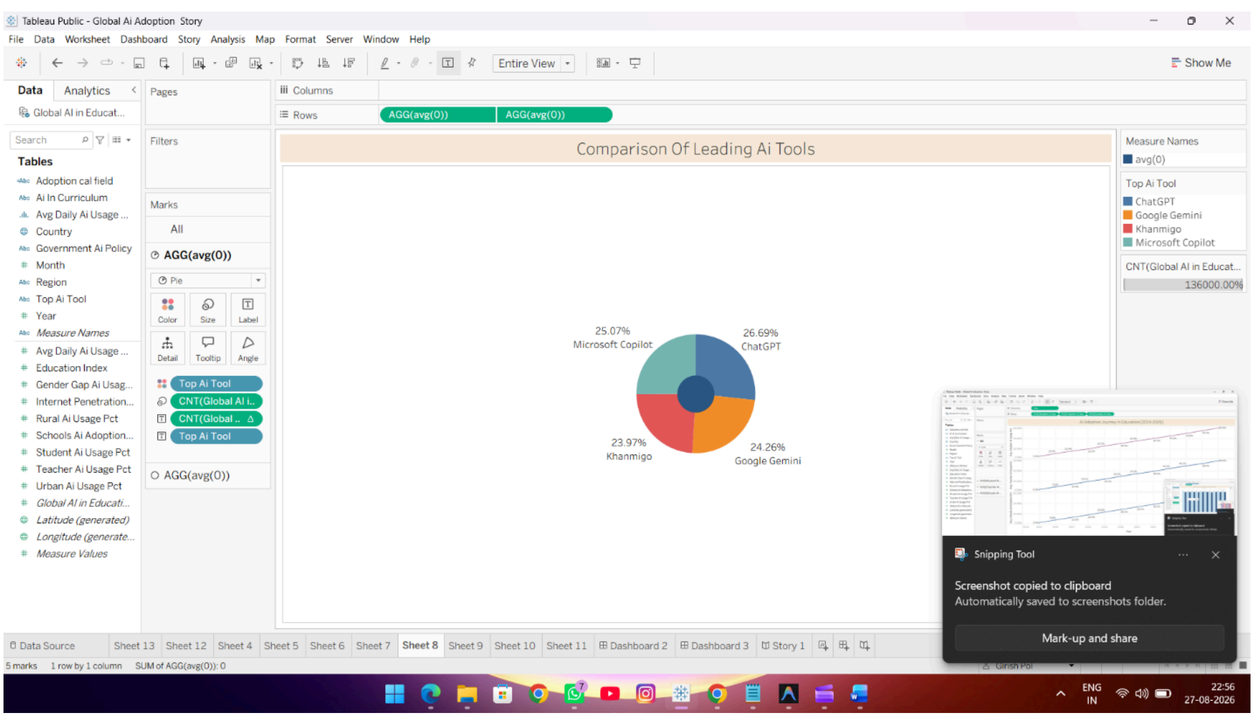
- **Top AI Tool** as the categorical dimension.
- **AI Tool Usage/Count** to represent the usage of each tool.
- AI tool categories for comparison.

The chart includes major AI tools such as **ChatGPT, Google Gemini, Khanmigo, and Microsoft Copilot**.

The visualization allows comparison of the representation and usage of different AI tools within the dataset. It helps identify which AI tools have greater adoption in the education sector.

Purpose:

This visualization helps users understand the **popularity and adoption of different AI tools in education** and provides a comparative view of the leading tools being used by educational users.



Visualization 6: Regional School AI Adoption

A **stacked bar chart** was developed to analyse school-level AI adoption across different geographical regions.

The visualization includes six major regions:

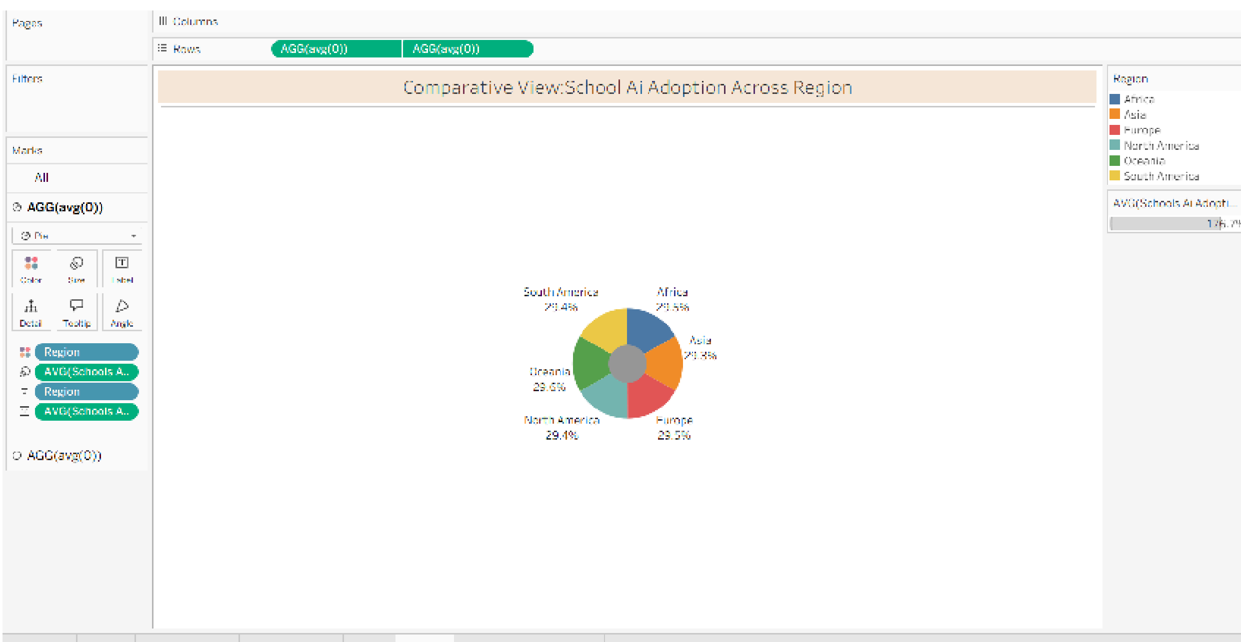
- **Africa**
- **Asia**
- **Europe**
- **North America**
- **Oceania**
- **South America**

The chart uses **school AI adoption** to compare the level of adoption across the different regions.

The visualization provides a comparative view of how schools across different parts of the world are adopting Artificial Intelligence. It helps identify regional differences and highlights variations in the level of AI integration within the education sector.

Purpose:

This visualization helps compare **school AI adoption across different regions** and provides insights into geographical differences in the implementation of Artificial Intelligence in education.



Visualization 7: AI Adoption Journey in Education (2014–2026)

A **line chart** was developed to analyse the growth of AI adoption in education over the period **2014 to 2026**.

The visualization uses:

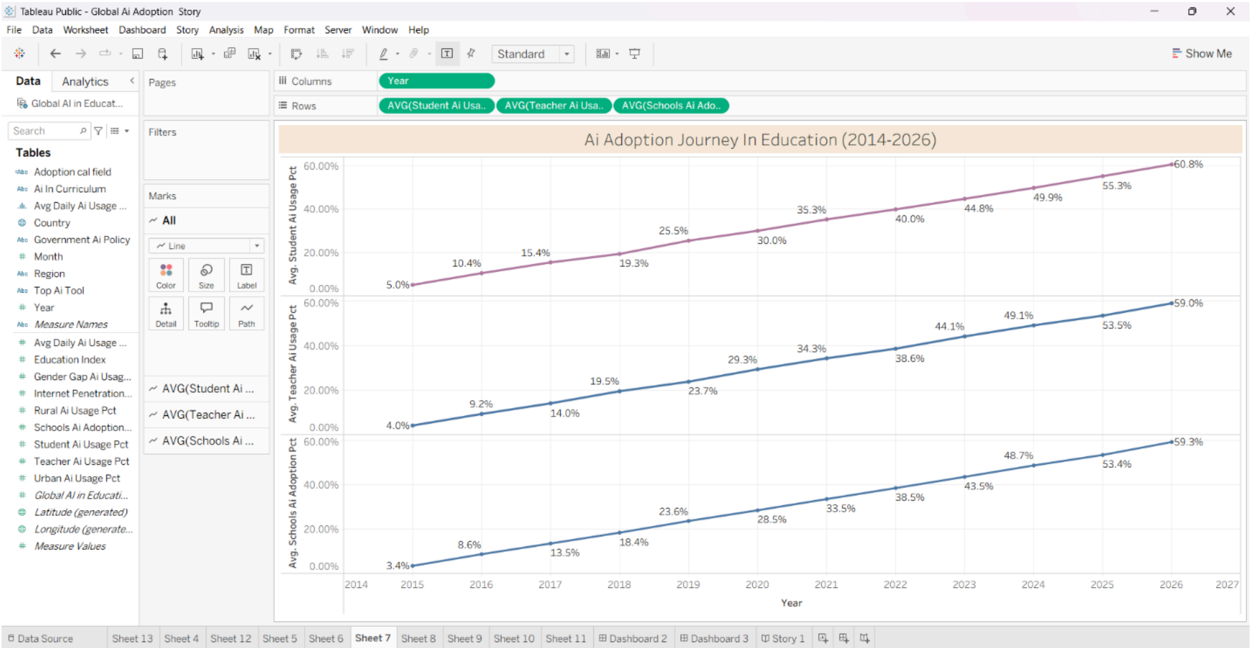
- **Year** as the time-based dimension.
- **Student AI Usage** to represent student adoption.
- **Teacher AI Usage** to represent teacher adoption.
- **School AI Adoption** to represent institutional adoption.

The visualization presents the changes in AI adoption across students, teachers, and schools over the selected period. The year-wise trend makes it easier to observe how the adoption of Artificial Intelligence has developed within the education sector over time.

The chart provides a comparative view of the adoption journey and highlights changes in AI usage among different educational groups.

Purpose:

This visualization helps identify **long-term trends in AI adoption from 2014 to 2026** and provides an understanding of how AI usage has evolved among students, teachers, and schools.



5. Dashboard

Dashboard Development

The individual visualizations were combined into an interactive **Tableau dashboard** titled:

"Global AI Adoption in Education"

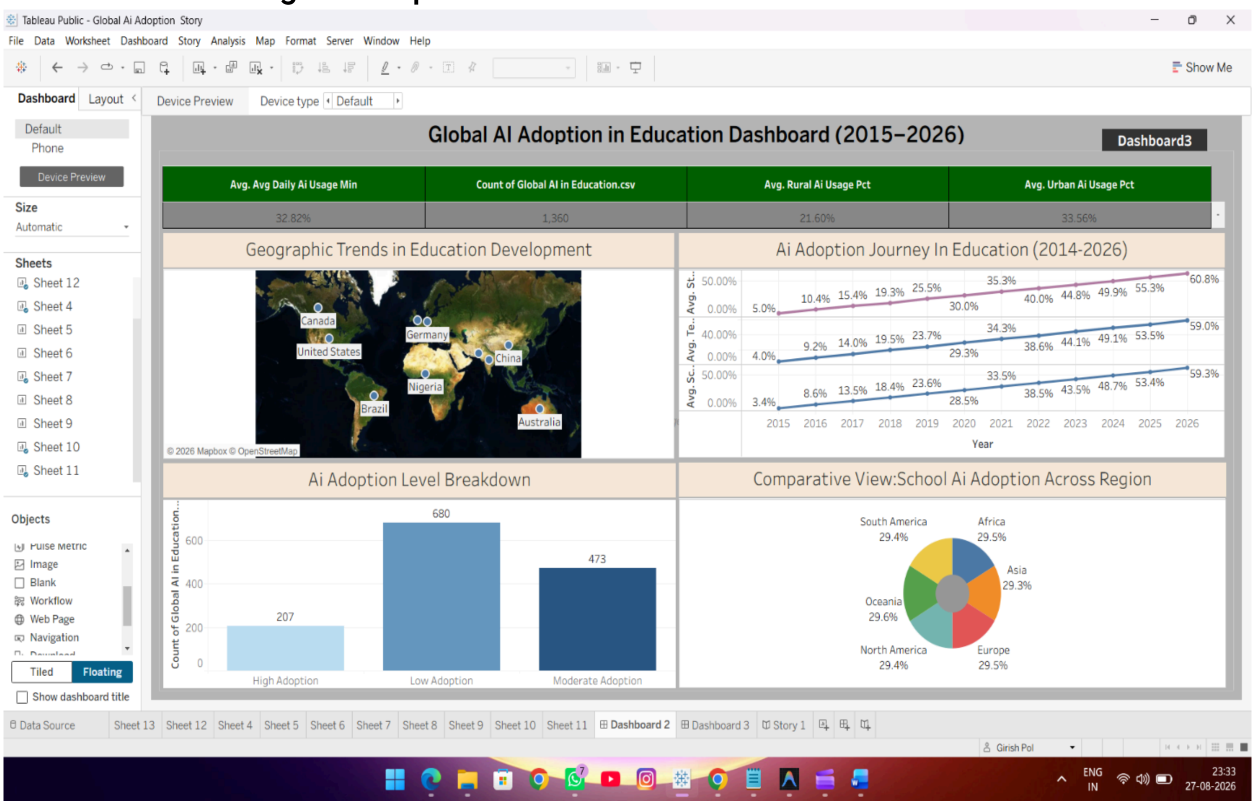
The dashboard integrates the major visualizations developed during the project into a single interface. It provides a comprehensive overview of AI adoption and usage in the education sector.

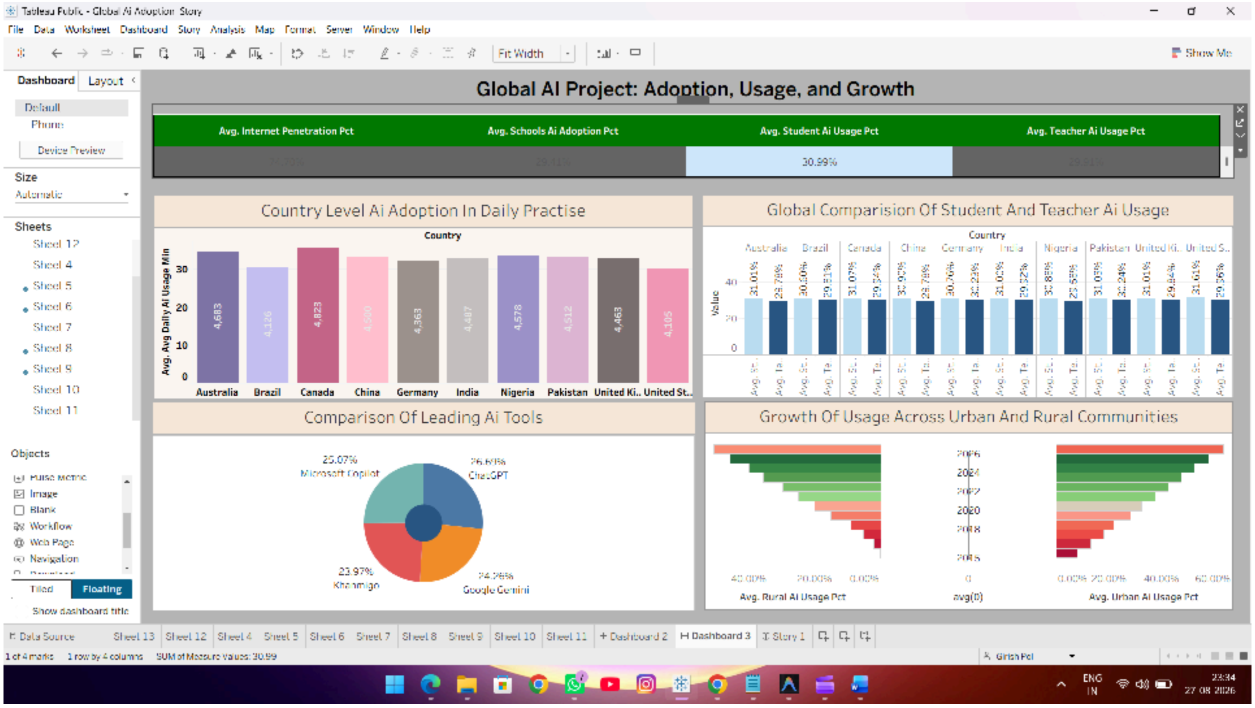
The dashboard contains:

Country/Region-based analysis
Average Daily AI Usage
Student AI Usage
Teacher AI Usage
Student vs Teacher AI Usage
AI Adoption Journey (2014–2026)
Top AI Tools Urban vs Rural AI Usage
School AI Adoption by Region
AI Adoption Level Analysis
Interactive filters
Geographical map
Charts and comparative visualizations

The dashboard allows users to interact with the data using relevant filters such as **Country, Region, Year, and AI adoption-related parameters**. Selecting different filter values updates the relevant visualizations and allows users to analyse the data from multiple perspectives.
This integrated design provides users with both a **high-level overview and detailed analysis of global AI adoption in education**.

Dashboard Title:Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report





6. Report

Tableau Story Development

A Tableau Story titled **“Global AI Adoption in Education”** was developed to present the visualizations in a sequential and understandable manner.
The story contains multiple story points that present different views of the Global AI in Education dataset.

The story includes:
1.**Geographic Trends in Education Development** – Shows the geographical distribution of AI adoption and education-related data across different countries.
2.**Country Level AI Adoption in Daily Practice** – Compares average daily AI usage across different countries.
3.**Student vs Teacher AI Usage** – Compares AI usage between students and teachers.
4.**AI Adoption Journey in Education (2014–2026)** – Shows the changes and growth of student, teacher, and school AI adoption over time.
5.**AI Tool Usage Comparison** – Compares the adoption and usage of leading AI tools such as ChatGPT, Google Gemini, Khanmigo, and Microsoft Copilot.
6.**Regional School AI Adoption** – Compares school-level AI adoption across different geographical regions.
7.**Urban vs Rural AI Usage** – Provides a comparative view of AI usage between urban and rural communities.
8.**AI Adoption Level Breakdown** – Presents the distribution of AI adoption across High, Moderate, and Low adoption levels.

The story format makes it easier to present the findings **step-by-step**, allowing users to understand the geographical, educational, regional, and temporal aspects of AI adoption in education.
Overall, the Tableau Story provides a structured representation of the project findings and supports a comprehensive understanding of **global AI adoption in the education sector**.

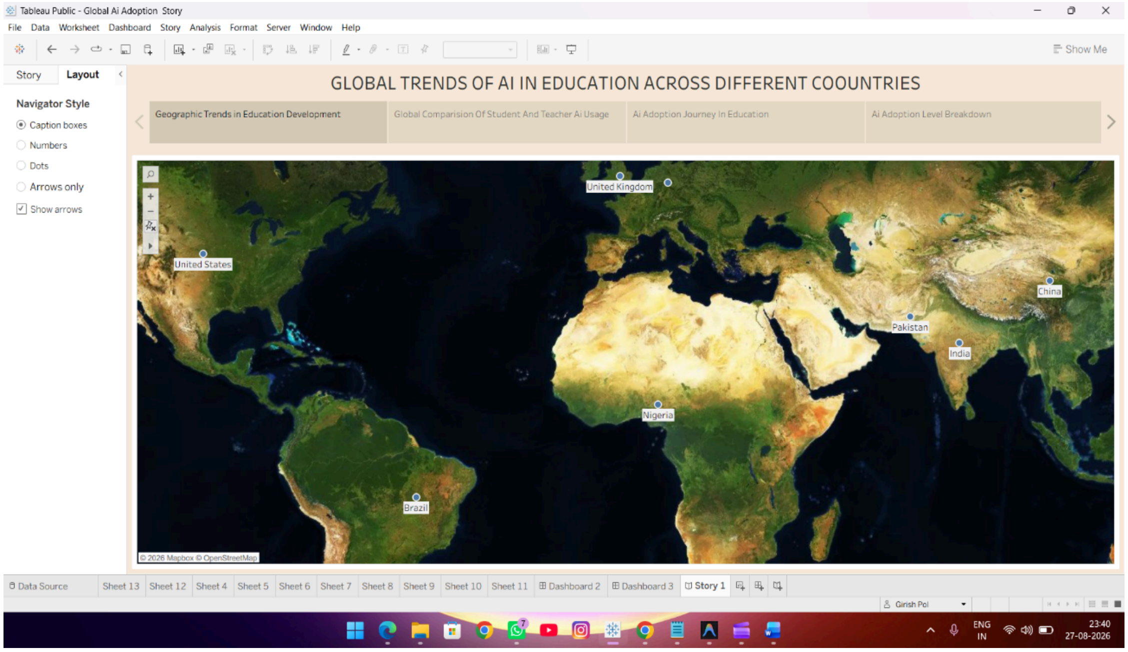


Tableau Public - Global Ai Adoption Story
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Ai Adoption Journey In Education
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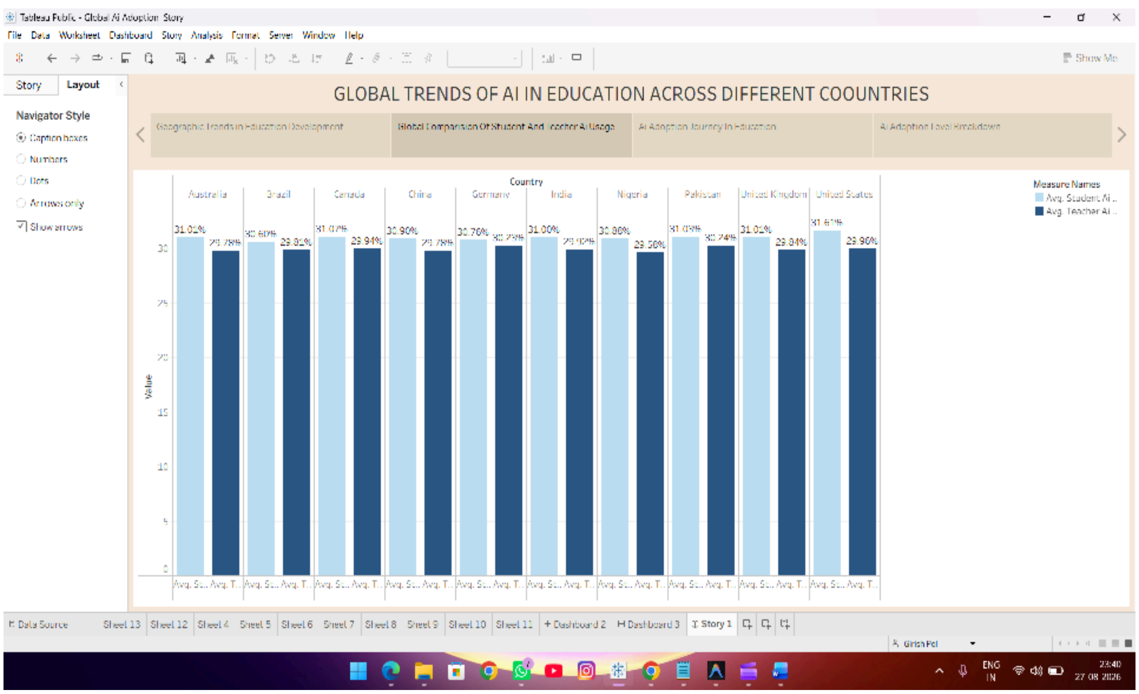
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Ai Adoption Journey In Education
Ai Adoption Level Breakdown
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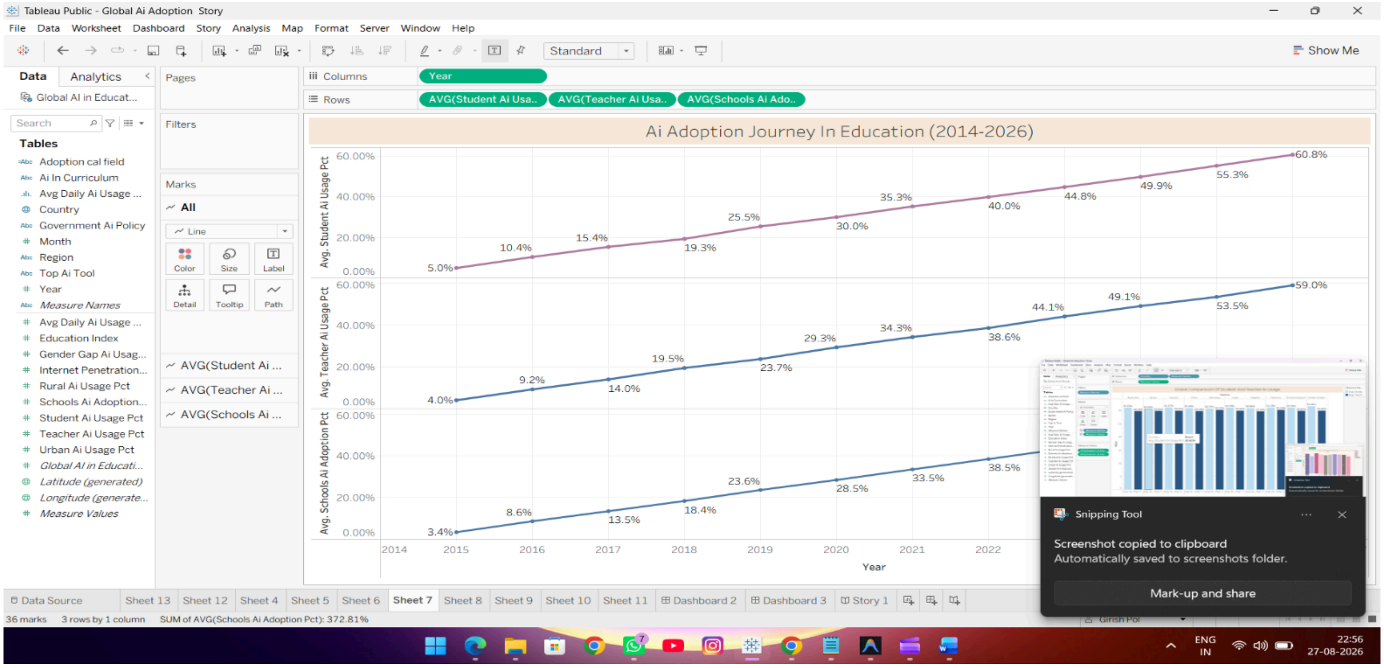
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Avg. Teacher Ai Usage Pct
4.0% 9.2% 14.0% 19.5% 23.7% 29.3% 34.3% 38.6% 44.1% 49.1% 53.5% 59.0%
Avg. Schools Ai Adoption Pct
3.4% 8.6% 13.5% 18.4% 23.6% 28.5% 33.5% 38.5%
Year
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Snipping Tool
Screenshot copied to clipboard
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Mark-up and share
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36 marks 3 rows by 1 column SUM of AVG(Schools Ai Adoption Pct): 372.81%
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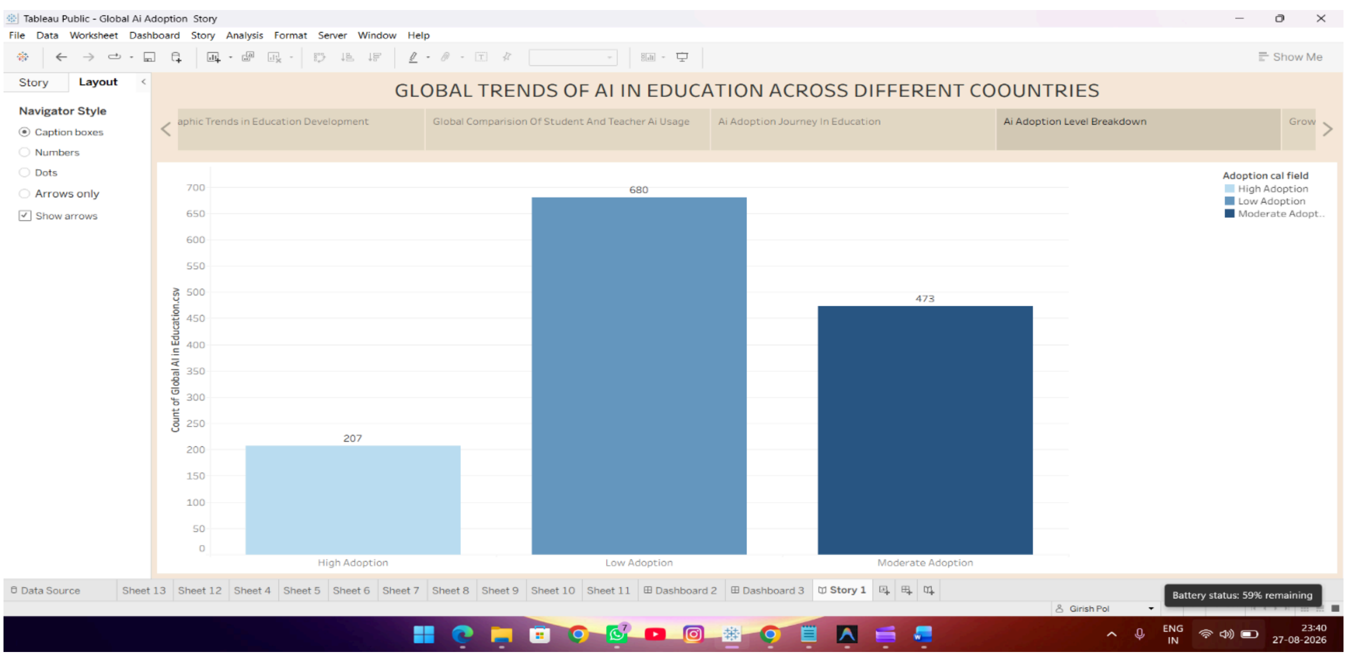
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Ai Adoption Level Breakdown
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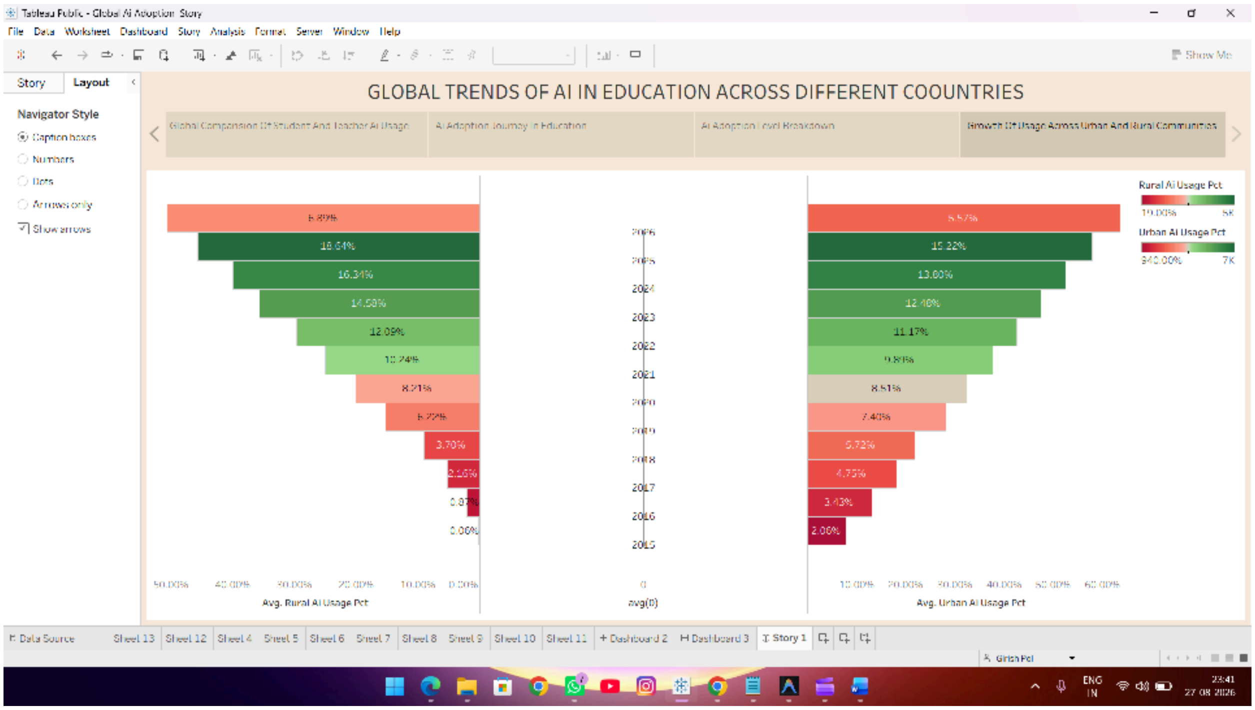

7. Additional Tableau Features

7.1. Utilization of Data Filters

Filters were used to allow users to interact with the dashboard and perform customized analysis.

The major filters include:

Country

Region

Year

AI Adoption Level

Top AI Tool

Other relevant AI adoption parameters available in the dashboard

When a user selects a filter, the relevant charts and visualizations update according to the selected criteria.

This functionality allows users to perform customized analysis without manually modifying the dataset.

Example

If the user selects a particular **Country**, the dashboard can display the corresponding AI usage, student and teacher AI adoption, school adoption, and other relevant educational AI indicators for that country.

7.2. Number of Calculated Fields

Calculated fields were used where required to derive additional insights from the available dataset.

Examples of calculated fields or derived measures include:

- Average Daily AI Usage
- Student AI Usage Percentage
- Teacher AI Usage Percentage
- School AI Adoption
- Urban AI Usage Percentage
- Rural AI Usage Percentage
- AI Adoption Level
- Regional AI Adoption
- Year-wise AI Adoption

Calculated Fields Used: As required by the dashboard analysis

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7.3. Multiple Visualizations

Multiple visualizations were developed to analyse **Global AI Adoption in Education**.

Examples include:

- KPI cards
- Geographic maps
- Bar charts
- Stacked bar charts
- Line charts
- Comparative charts
- AI tool analysis charts
- Regional adoption charts
- Urban vs Rural comparison charts
- Student vs Teacher comparison charts

Visualization Count

Total number of major visualizations: 8

8. Conclusion / Observation

The **Global AI Adoption in Education** project demonstrates how **Business Intelligence and data visualization** can be used to understand the rapidly growing adoption of Artificial Intelligence in the education sector.

The analysis provides insights into **country-wise AI usage, student and teacher AI usage, school AI adoption, AI adoption trends, leading AI tools, urban and rural AI usage, and regional adoption levels**.

From the **student perspective**, the dashboard helps understand the level of AI usage among learners and how student AI adoption has changed over time. It provides a clear view of the increasing role of AI tools in educational activities.

From the **teacher perspective**, the analysis highlights teacher AI usage and allows comparison between student and teacher adoption. This helps in understanding how educators are adopting AI alongside students.

From the **school and regional perspective**, the project analyses school-level AI adoption across different geographical regions and compares AI usage between **urban and rural communities**. These comparisons help identify differences in AI adoption across educational environments.

The analysis also provides insights into the adoption of leading AI tools such as **ChatGPT, Google Gemini, Khanmigo, and Microsoft Copilot**, along with the overall distribution of **High, Moderate, and Low AI Adoption** levels.

Overall, the project demonstrates that **Tableau can transform complex global education and AI adoption data into interactive and easy-to-understand dashboards**, helping users identify trends, compare regions and educational groups, and develop a better understanding of the role of Artificial Intelligence in modern education.

9. Future Scope

The **Global AI Adoption in Education** project can be enhanced in several ways in the future.

1. Real-Time Data Integration

The dashboard can be connected to continuously updated educational AI datasets or APIs to provide more current information about AI adoption and usage in the education sector.

2. Predictive Analytics

Machine learning models could be integrated to predict:

Future AI adoption trends in education

Student and teacher AI usage ****

School-level AI adoption

Emerging patterns in AI tool usage

Future regional AI adoption levels

3. AI Skill and Tool Demand Analysis

Future versions could analyse the skills and AI tools increasingly used in education, such as:

Generative AI

Machine Learning

Natural Language Processing
AI-assisted learning tools
AI-based assessment systems
Educational AI platforms

4. Advanced Geographic Analysis
AI adoption could be analysed at more detailed geographical levels, such as:
Country
State/Province
City
Region
This would help identify geographical areas with stronger or weaker AI adoption in education.

5. Personalized Educational Insights
A recommendation system could be developed to provide personalized insights based on a user's:
Student or teacher profile
AI usage level
Region
Educational requirements
Preferred AI tools
Learning or teaching needs

6. Advanced AI Integration

Generative AI could be integrated with the dashboard to automatically summarize visualization results, explain important trends, and provide personalized insights about AI adoption in education.

7. Automated Dashboard Updates

The project could be connected to automated data pipelines so that new educational AI data is periodically added and the Tableau dashboard is refreshed automatically.

Overall, these enhancements could transform the current visualization-based project into a more **intelligent, predictive, and interactive education analytics system**.

Appendix

10.1. GitHub & Project Demo Link

Project Demo Link https://drive.google.com/file/d/1M-32e1xFsYOvClCBd75NVdXh39MNE2H8/view?usp=sharing

GitHub Link –
https://github.com/omkardangare55-debug/AI-Career-Pulse-Decoding-the-Job-Market